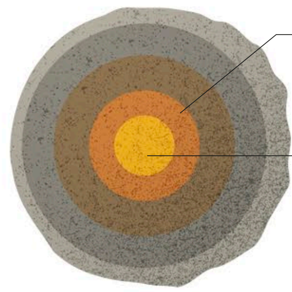
THE MOON

EARTH IS THE ONLY PLANET IN THE SOLAR SYSTEM WITH A MOON COMPARABLE IN SIZE TO ITSELF, BUT OUR NATURAL SATELLITE IS A COMPLETELY DRY, AIRLESS ENVIRONMENT.

Formation and structure

The Moon was almost certainly formed when a large, Mars-sized body struck a young Earth, throwing debris into space that eventually merged to form the body we see today. Its orbit has gradually widened and lengthened, and today, the Moon orbits Earth every 27.3 days. The Moon's interior was once warm enough for many volcanic eruptions, but such activity has now stopped.

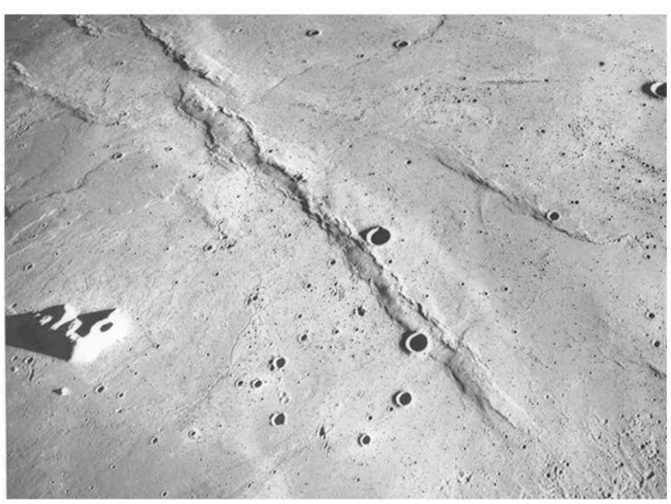


△ **Offset structure**
Tidal forces created by Earth's gravity early in the Moon's history distorted the Moon's symmetry.

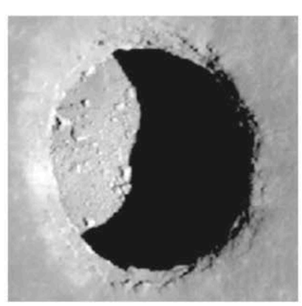
Surface features

The Moon's surface has been pounded by countless impacts for billions of years. Much of the ground is covered by craters of all sizes. Looking by eye from Earth, the obvious features on the Moon are its dark patches. These, the seas, or maria, are where very fluid lava flooded large impact basins. These large eruptions ended just over a billion years ago.

◁ **Hayn impact crater**
This typical impact crater was formed when an asteroid or comet hit the Moon. It has a flat floor and central peaks, where the Moon's surface rebounded.



◁ **Lava plain**
The maria are vast expanses of lava that have covered several large areas of the Moon. They are not, however, perfectly smooth. Wrinkle ridges show where the lava cooled and shrank.



◁ **Sink hole**
This pit, less than 330 ft (100 m) across, is where a sub-surface channel that once carried lava has collapsed. Similar features are seen in volcanic areas on Earth.